

Midpoint preservation under lens containment

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens_subset`

Applies the symmetric-lens midpoint theorem inside arbitrary source and target subsets.

Statement

Let $f: s \simeq t$ and $x, y \in s$. If the symmetric lens of radius r around x, y lies in s , the corresponding lens around $f(x), f(y)$ lies in t , and half the distance between x and y is at most r , then f sends $\text{midpoint}(x, y)$ to $\text{midpoint}(f(x), f(y))$.

Assumptions

Real normed vector spaces; f is an isometry equivalence s to t ; radius- r symmetric lenses about x, y and $f(x), f(y)$ lie in s and t ; half the distance from x to y is at most r .

Conclusion

f preserves the midpoint of x and y . The target half-distance bound follows from the isometry.

Proof idea

Restrict f and its inverse to the two contained lenses. Distance preservation transports the half-distance bound to the target foci. The lens-level midpoint theorem applied to this restricted isometry gives the result.

Proof steps

1. Construct the induced isometry equivalence between the two symmetric lenses.
2. Use `f.dist_eq` to identify the distance between the target foci with the distance between the source foci.
3. Transport the half-distance hypothesis to the target lens.
4. Apply midpoint preservation for an isometry between the two lenses.

Main citations

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens`

The proof or construction of Midpoint preservation under lens containment uses the project declaration “Midpoint preservation on a symmetric lens” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.IsometryEquiv.restrictToSymmetricLens`

The proof or construction of Midpoint preservation under lens containment uses the project declaration “Restriction of a set isometry to corresponding symmetric lenses” at the indicated step.

`MathlibAnnex.symmetricLens`

The proof or construction of Midpoint preservation under lens containment uses the project declaration “Symmetric lens” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.midpoint_mem_symmetricLens`

The proof or construction of Midpoint preservation under lens containment uses the project declaration “The midpoint belongs to a sufficiently large symmetric lens” at the indicated step.

Lean source signature (exact)

```
theorem map_midpoint_of_symmetricLens_subset
  {s : Set E} {t : Set F} (f : s  $\approx_i$  t) (x y : s) (r :  $\mathbb{R}$ )
  (hhalf :  $2^{-1} * \text{dist } (x : E) (y : E) \leq r$ )
  (hs : symmetricLens (x : E) (y : E) r  $\subseteq$  s)
  (ht : symmetricLens ((f x : t) : F) ((f y : t) : F) r  $\subseteq$  t) :
  ((f (midpoint  $\mathbb{R}$  (x : E) (y : E),
  hs (midpoint_mem_symmetricLens hhalf)) : t) : F) =
  midpoint  $\mathbb{R}$  ((f x : t) : F) ((f y : t) : F)
```

[Read exact MathlibAnnex v0.4.0 source](#)

Exact Card identity

Stable Card ID: e55f926ea1e3391b9a6426100e98fe7349c95254cdc2437a77d973137bea95c8

Card SHA-256: 8d98a636a7e168433d5141c38866083e4d514bbd9a686b436317cdeb3c85a05e

Approved exposition SHA-256: 09e97319542b53041fe9e0eba2c21ea36c537dc841dfcf8bc078456eef1c2cc2

AI-assisted material. Selected, edited, and reviewed by Ryotaro Tanaka. The recorded review is source-exposition correspondence, not an independent proof audit.